Obstruction Theorems for Farey-Spectral Approaches to the Riemann Hypothesis

Unit-Weight Nonexistence, Damped Trace Collapse, and Conditional Global-Trace No-Go Results

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Abstract

A spectral program can fail in two different ways. Its proposed operator may not exist on the intended domain, or the operator may exist while the statistic being read from it loses the arithmetic variation one hoped to recover. This paper proves both kinds of obstruction for a natural family built on the full Farey graph.

First, every Farey-graph vertex has infinite degree. With unit edge weights and arbitrary magnetic phases, the quadratic form is infinite on every nonzero finitely supported vector, so the proposed Laplacian does not exist over the standard core. Second, denominator damping by $w_e=(q_u q_v)^{-2\sigma}$ creates a sharp real-exponent existence boundary: the positive operator exists for $\sigma>1/2$, fails over the core for $\sigma\leq 1/2$, and is trace class in the admissible half-line with $\operatorname{tr} L(\sigma)=2\zeta(2\sigma)^2/\zeta(4\sigma)$.

The exact trace identity produces a third result: spectral mass collapses toward zero as the Farey truncation grows. For every fixed Lipschitz test function, the global trace has the rigid form $g(0)m_N+O(1)$; for every fixed continuous test function its normalized trace tends to $g(0)$. A self-contained Farey-counting argument proves the needed unconditional discrepancy estimate $D(N)=O(N\log N)=o(m_N)$. It follows unconditionally that any proposed affine trace identity must cancel its extensive term. Complete nonexistence in the remaining bounded or sub-power branch is stated only as a conditional corollary under separately named growth hypotheses; those hypotheses are not claimed as results of this paper.

The paper therefore supplies an operator-existence obstruction, an exact trace-collapse theorem, and a lossless map of what remains open. It makes no claim to prove the Riemann Hypothesis or full physical Super Information Theory.

Scope and evidence boundary

 Evidence classes. **[proved]** denotes a self-contained theorem or an exactly cited result. **[measured]** denotes finite computation used by no theorem. **[conjectural]** denotes a proposed direction used by no theorem. An explicitly conditional corollary is a proved implication whose additional hypothesis is not asserted as an established fact.

1. Introduction

1.1 The problem before the theorem names

The ordinary question is whether a graph can be turned into a useful spectral instrument. One first assigns weights to edges, builds a Laplacian, and asks whether its eigenvalues or traces retain the number-theoretic structure encoded by the graph.

Two checks must come before interpretation. First, does the infinite-graph operator exist on the domain where graph Laplacians are normally defined? Second, if damping makes it exist, does a global spectral average retain enough variation to reproduce the target arithmetic observable?

For the full Farey graph, the first check fails at unit weight because every vertex has infinitely many neighbors. Denominator damping repairs existence, but the repair has a consequence: total spectral mass becomes finite while the number of finite-truncation eigenvalues grows quadratically. The spectrum therefore collapses toward zero in its empirical distribution. Only after those operations are understood do we name the results: the unit-weight domain obstruction, the critical-exponent existence dichotomy, and global-trace rigidity.

The target observable is the Franel-Landau discrepancy $D(N)=\Sigma_j\ |\rho_j-j/m_N|$ of the Farey sequence F_N. Franel and Landau connected its fine asymptotic scale to the Riemann Hypothesis. This paper does not prove that fine scale. It proves what the proposed operator and its global traces can and cannot do.

- **Theorem A (unit-weight domain obstruction) [proved].** Unit edge weights, with arbitrary magnetic phases, give infinite quadratic-form energy to every nonzero finitely supported vector. No Laplacian of the stated form exists over the standard core.
- **Theorem B (damped-family existence and exact trace) [proved].** For $w_e=(q_uq_v)^{-2\sigma}$, the positive operator exists exactly for real $\sigma>1/2$, is trace class there, and has $\mathrm{tr}\ L(\sigma)=2\zeta(2\sigma)^2/\zeta(4\sigma)$.
- **Theorem C (global-trace rigidity) [proved].** For $\sigma>1/2$, fixed Lipschitz tests satisfy $\mathrm{tr}\,g(L_N)=g(0)\, m_N+O(1)$, while fixed continuous tests satisfy $\mathrm{tr}\,g(L_N)/m_N\to g(0)$. The paper proves directly that $D(N)=O(N\log N)=o(m_N)$, so every affine trace identity must cancel its extensive term.
- **Conditional exclusion corollaries [proved implications].** If one additionally assumes that $D(N)$ is unbounded, Lipschitz affine global-trace identities are impossible. If one assumes the displayed subsequential lower bound, the same conclusion holds for Hölder exponents above $3/4$. These extra hypotheses are not promoted as established results here.

The former shell-localization measurements are preserved only as historical route information because their promised reproducibility package was not available. Distinguished-vector and cusp observables remain outside the global-trace fence.

1.2 Why publish obstructions

Negative results that delimit a search space are rarely published in this area; we record ours with proofs and explicit input contracts. Each theorem removes only the family named in its hypotheses. Former measured exclusions are omitted rather than promoted. Anyone building spectral operators on Farey / Stern-Brocot structure can start where this paper ends.

There is precedent for publishing limitative and reformulation results around RH. The Nyman-Beurling closure criterion (Nyman 1950; Beurling 1955) and its strengthening by Báez-Duarte (2003) recast RH as a Hilbert-space approximation problem, and Burnol (2002) proved an unconditional lower bound - a published *floor* - on the relevant distance. Jorgensen (2008) established essential self-adjointness for a large class of conductance-weighted graph Laplacians on the finitely supported domain; Section 2 explains why the unit-weight Farey expression considered here fails even to define such an operator on that domain.

1.3 Related work

On the Farey / Stern-Brocot side specifically, the spectral literature our obstructions are aimed at includes the Farey fraction spin chain of Kleban-Özlük (1999) and Knauf's number-theoretic chains, the Farey-map transfer operators of Prellberg and of Bonanno-Isola (2008), the fine statistics of Farey fractions of Boca-Cobeli-Zaharescu, and the unweighted Farey-graph Laplacian spectra studied as small-world networks by Zhang-Comellas (2011). None of these works builds a weighted (magnetic) Laplacian on the Farey graph with arithmetic weights; the theorems below show precisely which such constructions cannot work, and the natural construction at unit weights is fenced first. The transfer-operator results of Prellberg and Bonanno-Isola concern a different operator class (composition / transfer operators of the Farey map, not graph Laplacians). The former finite comparison is not retained as a result in v7.

1.4 Roadmap

Section 2 defines the Farey graph and proves the unit-weight domain obstruction. Section 3 proves the damped-family existence dichotomy, the exact trace identity, spectral collapse, a self-contained sublinear discrepancy bound, and the global-trace rigidity theorem with its explicitly conditional corollaries. Section 4 records why the earlier measured exclusions are not retained. Section 5 identifies observable classes that survive. Section 6 states discriminating open problems.

2. Setting and nonexistence at unit weights

Definition 2.1 (Farey graph, spaces, discrepancy). (a) The *full Farey graph* $\mathfrak{F}$ has vertex set the reduced fractions $p/q \in [0, 1]$ (with $0 = 0/1$, $1 = 1/1$), and an edge $\{p/q, r/s\}$ exactly when $|ps - qr| = 1$ (*Farey adjacency*; geometrically, tangency of the corresponding Ford circles, Ford 1938). The endpoint convention - $0/1 \sim 1/1$ is an edge, since $|0\cdot 1 - 1\cdot 1| = 1$ - follows companion Paper 3, Appendix A, and is used identically throughout the series. (b) $\ell^2(\mathfrak{F})$ is the Hilbert space of square-summable functions on the vertex set; $c\_c(\mathfrak{F}) \subset \ell^2(\mathfrak{F})$ is the dense subspace of finitely supported vectors (the *core*). (c) F_N is the Farey sequence of order N (reduced fractions in $[0, 1]$ with denominator $q \leq N$), $m\_N := |F\_N|$. Classically $m\_N = 3N^2/\pi^2 + O(N \log N)$. (d) A *magnetic weight assignment* attaches to each edge $e = \{u, v\}$ a weight $w\_e > 0$ and a phase $\theta\_e \in \mathbb{R}$. The associated quadratic form on $c\_c(\mathfrak{F})$ is $Q(\psi) := \Sigma\_e\ w\_e\ |\psi(u) - e^{\{i\theta\_e\}}\ \psi(v)|^2$, equivalently $\|A\psi\|^2$ for the weighted, phased incidence operator A. When all $\theta\_e = 0$ and the form is bounded, the associated self-adjoint operator is the weighted graph Laplacian L. (e) The *Franel-Landau discrepancy* is $D(N) := \Sigma\_\{j=1\}^\{m\_N\}\ |\rho\_j - j/m\_N|$, where $\rho\_1 < \cdots < \rho\_\{m\_N\}$ enumerates F_N. (f) The evidence-class convention of the status block applies: [proved] / [measured] / [conjectural]. Nothing labeled [measured] or [conjectural] is used by, or supports, any theorem in this paper.

Lemma 2.2 (infinite degree) [proved]. Every vertex of $\mathfrak{F}$ has infinite degree. Precisely: for reduced p/q with a Bézout solution $p\ s_0 - q\ r_0 = \varepsilon \in \{\pm 1\}$, $1 \leq s_0 \leq q$, the family $v\_t := (r_0 + tp)/(s_0 + tq)$, $t \geq 0$, consists of pairwise distinct reduced Farey neighbors of p/q. Moreover, the set of neighbor denominators of p/q is exactly $\{s \geq 1 : s \equiv s_0 \pmod q\} \cup \{s \geq 1 : s \equiv q - s_0 \pmod q\}$ (two arithmetic progressions of common difference q), each denominator realized by exactly one neighbor on each side. In particular, for $q \geq 2$ the vertex p/q has exactly two neighbors of denominator $< q$, namely those with denominators s_0 and $q - s_0$ (its Stern-Brocot parents).

Proof. For every $t \geq 0$, $p(s_0 + tq) - q(r_0 + tp) = p\ s_0 - q\ r_0 = \varepsilon$, so $|p(s_0+tq) - q(r_0+tp)| = 1$: each v_t is Farey-adjacent to p/q, and any common divisor of $r_0 + tp$ and $s_0 + tq$ would divide $\varepsilon = \pm 1$, so each v_t is reduced. The denominators $s_0 + tq$ strictly increase in t (as $q \geq 1$), so the v_t are pairwise distinct; the degree of p/q is therefore infinite. For the converse description: if $r/s \sim p/q$ then $ps - qr = \pm 1$, so $s \equiv \pm s_0 \pmod q$ according to the sign; conversely for each s in either progression the numerator r is uniquely determined by the determinant equation and $0 \leq r/s \leq 1$ (for the endpoint vertices $0/1$ and $1/1$, restrict to $t \geq 1$ where needed to stay in $[0, 1]$; the families remain infinite, e.g. the neighbors $1/(t+1)$ of $0/1$). The parent statement is the case $t = 0$ of each progression:

for $q \geq 2$ both s_0 and $q - s_0$ lie in $\{1, \dots, q - 1\}$, each realized by exactly one neighbor, and every neighbor denominator $< q$ is a $t = 0$ term. ■

Theorem 2.3 (nonexistence at unit weights) [proved]. Let $w_e \equiv 1$ and let (θ_e) be an arbitrary phase assignment. Then $Q(\psi) = +\infty$ for every $\psi \in c_c(\mathfrak{F}) \setminus \{0\}$. Consequently the form domain of Q intersects $c_c(\mathfrak{F})$ only in $\{0\}$, and no self-adjoint operator of the form $V + A^\dagger A$ (any diagonal V) exists over the finitely-supported core in $\ell^2(\mathfrak{F})$.

Proof. Let $\psi \in c_c(\mathfrak{F})$, $\psi \neq 0$, and pick v with $\psi(v) \neq 0$. By Lemma 2.2, v has infinitely many neighbors; all but finitely many lie outside the finite support of ψ. For each such neighbor u ($\psi(u) = 0$), the edge $e = \{v, u\}$ contributes $w_e\ |\psi(v) - e^{i\theta_e} \psi(u)|^2 = |\psi(v)|^2 > 0$ to $Q(\psi)$ - the phase multiplies the vanishing term and is irrelevant since $|e^{i\theta_e}| = 1$. Infinitely many contributions, each equal to $|\psi(v)|^2$, give $Q(\psi) = +\infty$. A diagonal V changes Q by a finite sum on $c_c(\mathfrak{F})$ and cannot repair divergence. ■

Remark 2.4 (domain scope). The proof shows more than failure on the core: any vector ψ of finite form energy must satisfy $\psi(v) = 0$ at every vertex v for which infinitely many neighbors u have $\psi(u) = 0$. In particular the form domain contains no finitely supported vector other than 0. We do not rule out exotic dense domains of nowhere-vanishing vectors with delicately correlated tails (Question Q4, §6); the claim proved is nonexistence over the core, which is the canonical domain on which graph Laplacians are built (cf. Jorgensen 2008).

Remark 2.5 (failure mode and handoff). Theorem 2.3 identifies the precise failure of the natural "adjacency operator on all Farey fractions" construction: the object is not an operator. Any repair must damp edges. Quantitatively, by Lemma 2.2 the neighbor denominators of p/q form two arithmetic progressions of common difference q - density $2/q$ per unit denominator - and it is exactly these progressions that §3 re-weights by $(q_u q_v)^{-2\sigma}$ and resums as Hurwitz-zeta tails: the §2→§3 handoff is structural. Theorem 3.1 shows damping below the critical exponent is impossible, and Lemma 3.2.2 + Theorem 3.4 show damping above it is fatal to global-trace signals.

Remark 2.6 (provenance). Theorem 2.3 restates Theorem C-3 of proof packet BC-001-proof-packet-20260612 (archived); the statements match.

3. The critical dichotomy and global-trace rigidity

Standing notation for §3. The spectral parameter σ is real. $L(\sigma)$ denotes the graph Laplacian on $\ell^2(\mathfrak{F})$ with positive edge weights $w_e=(q_u q_v)^{-2\sigma}$ and zero phases; $L_N(\sigma)$ is the corresponding finite induced-graph Laplacian on F_N, extended by zero on its orthogonal complement. The phrase "critical line" records that σ is the real part of the complex variable used in zeta-function applications; this paper does not claim positivity or self-adjointness for a complex-weight operator. The weighted degree of $v=p/q$ is $W_\sigma(v):=\Sigma_{u\sim v}(q\ q_u)^{-2\sigma}$, and $W_\sigma^{(N)}(v)$ restricts that sum to neighbors in F_N. For $a>0$, $\zeta(z,a)=\Sigma_{t\geq 0}(t+a)^{-z}$ is the Hurwitz zeta function.

3.1 Existence: the dichotomy

Lemma 3.0 (weighted-degree bound, with constants) [proved]. For $\sigma>1/2$ and an interior vertex $v=p/q$, $q\geq 2$, $W_\sigma(v)=q^{-4\sigma}[\zeta(2\sigma,s_0/q)+\zeta(2\sigma,(q-s_0)/q)]\ \leq 2\zeta(2\sigma)q^{-2\sigma}\leq 2\zeta(2\sigma)$, with s_0 as in Lemma 2.2. For either endpoint $q=1$, $W_\sigma(v)=\zeta(2\sigma)$, so the same upper bound holds.

Proof. By Lemma 2.2 the neighbor denominators are $\{a + tq : t \geq 0\}$ for $a \in \{s_0, q - s_0\}$ (each realized once). Hence $W_\sigma(v) = q^{-2\sigma} \Sigma_a \Sigma_{t\geq 0} (a + tq)^{-2\sigma} = q^{-2\sigma}$

$\Sigma\_a\ q^{-2\sigma}\ \zeta(2\sigma,\ a/q) = q^{-4\sigma}\ \Sigma\_a\ \zeta(2\sigma,\ a/q)$. For an interior vertex, both residues satisfy $1 \le a \le q-1$. For $0<x\le 1$, $\zeta(2\sigma,x)=x^{-2\sigma}+\Sigma\_{t\ge 1}(t+x)^{-2\sigma} \le x^{-2\sigma}+\zeta(2\sigma)$. With $x=a/q \ge 1/q$ this gives $q^{-4\sigma}\ \zeta(2\sigma,\ a/q) \le q^{-4\sigma}(\ (q/a)^{2\sigma} + \zeta(2\sigma)\ ) \le q^{-2\sigma} + q^{-4\sigma}\zeta(2\sigma) \le \zeta(2\sigma)\ q^{-2\sigma}$ (using $\zeta(2\sigma) \ge 1$ and $q \ge 1$). Summing over the two values of a yields the interior bound. For $0/1$, the neighbors are exactly $1/n$, $n\ge 1$; for $1/1$ they are $(n-1)/n$, $n\ge 1$. Thus each endpoint has weighted degree $\Sigma\_{n\ge 1}n^{-2\sigma}=\zeta(2\sigma)$. ■

Theorem 3.1 (critical-exponent existence dichotomy) [proved]. Fix the real positive-weight family $w\_e=(q\_uq\_v)^{-2\sigma}$ on $\mathfrak{F}$. (a) If $\sigma > 1/2$, the quadratic form is bounded on $\ell^2(\mathfrak{F})$ and the associated Laplacian satisfies $0 \preceq L(\sigma) \preceq 4\zeta(2\sigma)\cdot I$; in particular $\|L(\sigma)\| \le 4\zeta(2\sigma)$. (b) If $\sigma \le 1/2$, then $Q(\psi) = +\infty$ for every $\psi \in c\_c(\mathfrak{F}) \setminus \{0\}$, and no operator exists over the core. The real convergence boundary is exactly $\sigma=1/2$, corresponding to the real part of the zeta critical line.

Proof. (a) For $\psi \in c\_c(\mathfrak{F})$, the elementary inequality $|\psi(u) - \psi(v)|^2 \le 2|\psi(u)|^2 + 2|\psi(v)|^2$ gives $Q(\psi) = \Sigma\_e\ w\_e\ |\psi(u) - \psi(v)|^2 \le 2\ \Sigma\_v\ W\_\sigma(v)\ |\psi(v)|^2 \le 4\zeta(2\sigma)\ \|\psi\|^2$ by Lemma 3.0. The form is densely defined, nonnegative, and bounded, so it extends to all of $\ell^2(\mathfrak{F})$ and defines a bounded self-adjoint $L(\sigma)$ with $0 \preceq L(\sigma) \preceq 4\zeta(2\sigma)\cdot I$. (b) For $\sigma \le 1/2$ and $a \in \{s_0,\ q - s_0\}$, the progression sum $\Sigma\_\{t\ge 0\}\ (a + tq)^{-2\sigma}$ diverges (exponent $2\sigma \le 1$, the harmonic threshold), so every vertex has infinite weighted degree. The argument of Theorem 2.3 applies verbatim with weights: each edge from v to the complement of the support contributes $w\_e\ |\psi(v)|^2$, and these weights sum to $+\infty$. ■

Remark 3.2 (boundary; constants discharged). (a) The convergence threshold aligns with the real part of the line central to RH: the positive operator exists precisely for $\sigma>1/2$ and degenerates as $\sigma \downarrow 1/2$. Delocalization evidence at the boundary is reported in packet BC-017-critical-descent-results-20260612 ([measured]; not used here). The former pending Ford-circle illustration is omitted because it is not part of the proof or evidence package. (b) The norm constant $4\zeta(2\sigma)$ is derived self-contained in Lemma 3.0 and Theorem 3.1(a) (per-vertex bound $2\zeta(2\sigma)$, quadratic-form doubling gives the factor 2); the "factor bookkeeping" flag in packet BC-016-fordfamily-zeta-norm-20260612 is thereby discharged for this bound. The *exact* Hurwitz-zeta diagonal evaluation (Lemma 3.0's first equality, summed) is not needed in this paper and remains with companion Paper 7.

3.2 Finite-N spectral structure

Lemma 3.2.1 (strong convergence, general $\sigma > 1/2$) [proved]. $L\_N(\sigma) \to L(\sigma)$ strongly on $\ell^2(\mathfrak{F})$ as $N \to \infty$.

Proof. Let $\psi \in c\_c(\mathfrak{F})$ with support S, and N large enough that $S \subset F\_N$. Then $(L(\sigma) - L\_N(\sigma))\psi$ involves only edges with one endpoint in S and the other of denominator $> N$. For $v = p/q \in S$, $\|(L(\sigma) - L\_N(\sigma))\psi\| \le \Sigma\_\{v\in S\}\ |\psi(v)|\ \cdot 2 \cdot$ (tail of $W\_\sigma(v)$ beyond denominator N), and the tail $q^{-2\sigma}\ \Sigma\_a\ \Sigma\_\{a+tq > N\}\ (a + tq)^{-2\sigma} \le 2\ q^{-2\sigma}\ \Sigma\_\{n > N\}\ n^{-2\sigma} \to 0$ as $N \to \infty$, since $2\sigma > 1$. So $L\_N(\sigma)\psi \to L(\sigma)\psi$ on the dense core; the uniform bound $\|L\_N(\sigma)\| \le 4\zeta(2\sigma)$ (Theorem 3.1(a) applied to the induced graph, whose weighted degrees are dominated by W_σ) extends strong convergence to all of $\ell^2(\mathfrak{F})$ by an $\varepsilon/3$ argument. (This mirrors the $\sigma = 1$ proof in packet BC-001/companion Paper 1.) ■

Lemma 3.2.2 (exact trace identity; uniform trace bound) [proved]. Let $\sigma > 1/2$ and $T\_N(\sigma) := \mathrm{tr}\ L\_N(\sigma) = \Sigma\_\{v\in F\_N\}\ W\_\sigma^{(N)}(v)$. Then $L(\sigma)$ is trace class, and $\mathrm{tr}\ L(\sigma) = 2\ \Sigma\_\{\text{edges } \{u,v\} \text{ of } \mathfrak{F}\}\ (q\_u\ q\_v)^{-2\sigma} = 2\ \zeta(2\sigma)^2\ /\ \zeta(4\sigma)$. Consequently $T\_N(\sigma)$ is nondecreasing in N with $T\_N(\sigma) \le T(\sigma) := 2\zeta(2\sigma)^2/\zeta(4\sigma) < \infty$ for all N, and $T\_N(\sigma) \to T(\sigma)$ as $N \to \infty$.

Proof. The trace of a weighted graph Laplacian is the sum of weighted degrees, i.e. twice the sum of edge weights; all terms are positive, so monotonicity in N and the limit statement follow from the edge-sum evaluation, and trace-class-ness of the positive operator $L(\sigma)$ follows from finiteness of the supremum of $\mathrm{tr}\ P\ L(\sigma)\ P$ over finite-rank coordinate projections P, which is the same edge sum.

It remains to evaluate $\Sigma_{\{edges\}}\ (q_u\ q_v)^{-2\sigma}$. Orient each edge toward its endpoint of strictly larger denominator when the denominators differ; by Lemma 2.2 (parent statement), each vertex p/q with $q \geq 2$ is the head of exactly two edges, whose tails have denominators $s_0(p,\ q)$ and $q - s_0(p,\ q)$, both in $\{1,\ \dots,\ q - 1\}$. The only edge with equal endpoint denominators is $0/1 \sim 1/1$ (an edge $\{p/q,\ r/q\}$ with $q \geq 2$ would force $q \mid (ps - qr) = \pm 1$), and it contributes $(1\cdot 1)^{-2\sigma} = 1$. As p runs over the $\phi(q)$ residues coprime to q, the map $p \mapsto s_0(p,\ q)$ (the representative of $\pm p^{-1} \bmod q$ selected by the Bézout normalization) is a bijection onto the residues in $\{1,\ \dots,\ q - 1\}$ coprime to q (for $q \geq 2$), and $p \mapsto q - s_0(p,\ q)$ is likewise a bijection onto the same set. Hence $\Sigma_{\{edges\}}\ (q_u\ q_v)^{-2\sigma} = 1 + \Sigma_{\{q \geq 2\}}\ q^{-2\sigma}\ \Sigma_{\{p\ coprime\}}\ [\ s_0^{-2\sigma} + (q - s_0)^{-2\sigma}\ ] \quad = 1 + 2\ \Sigma_{\{q \geq 2\}}\ q^{-2\sigma}\ \Sigma_{\{1 \leq a < q,\ (a,q)=1\}}\ a^{-2\sigma} \quad = 1 + 2\ \Sigma_{\{(a,q):\ (a,q)=1,\ q > a \geq 1\}}\ (aq)^{-2\sigma}$. Now classify ordered coprime pairs $(m,\ n)$ by trichotomy: $m = n$ forces $m = n = 1$; the pairs with $n > m$ and with $m > n$ contribute equally. Since $\Sigma_{\{(m,n)\ coprime\}}\ (mn)^{-2\sigma} = \zeta(2\sigma)^2/\zeta(4\sigma)$ (standard Möbius inversion: $\Sigma_{\{m,n\}}\ (mn)^{-2\sigma}\ \Sigma_{\{d \mid (m,n)\}}\ \mu(d) = \Sigma_d\ \mu(d)\ d^{-4\sigma}\ \zeta(2\sigma)^2 = \zeta(2\sigma)^2/\zeta(4\sigma)$), we get $\Sigma_{\{q > a \geq 1,\ coprime\}}\ (aq)^{-2\sigma} = (\zeta(2\sigma)^2/\zeta(4\sigma) - 1)/2$, so $\Sigma_{\{edges\}}\ (q_u\ q_v)^{-2\sigma} = 1 + (\zeta(2\sigma)^2/\zeta(4\sigma) - 1) = \zeta(2\sigma)^2/\zeta(4\sigma)$, and $\mathrm{tr}\ L(\sigma) = 2\zeta(2\sigma)^2/\zeta(4\sigma)$. ■

Corollary 3.2.3 (spectral collapse) [proved]. Let $\sigma > 1/2$ and $\varepsilon > 0$. Then, uniformly in N, $\#\{i : \lambda_i(L_N(\sigma)) \geq \varepsilon\} \leq T_N(\sigma)/\varepsilon \leq T(\sigma)/\varepsilon = O_\sigma(1/\varepsilon)$, and since $m_N \asymp N^2$, the empirical spectral distribution $(1/m_N)\ \Sigma_i\ \delta_{\{\lambda_i(L_N)\}}$ converges weakly to δ_0 as $N \to \infty$.

Proof. Markov's inequality on the nonnegative spectrum gives the counting bound, with $T_N(\sigma) \leq T(\sigma)$ by Lemma 3.2.2; weak convergence to δ_0 follows since for any bounded continuous f, $(1/m_N)|\Sigma_i\ f(\lambda_i) - m_N\ f(0)| \leq \omega_f(\varepsilon) + 2\ \sup|f|\ \cdot\ T(\sigma)/(\varepsilon\ m_N) \to 0$ on letting $N \to \infty$ then $\varepsilon \to 0$. ■

Remark 3.2.4 (correction of earlier drafts; numerical check). Drafts v1-v4 of this paper claimed two-sided growth $T_N(\sigma) \asymp N^{2-2\sigma}$ for $1/2 < \sigma < 1$ and $T_N(1) \asymp \log N$. The upper bounds were correct but lossy (they summed the per-vertex bound $2\zeta(2\sigma)\ q^{-2\sigma}$ of Lemma 3.0 over all $\phi(q)$ vertices of denominator q, whereas $W_\sigma(p/q)$ is of that size only for the few p with small Bézout residue s_0); the claimed lower bounds were false. Lemma 3.2.2 replaces them with the exact identity. Numerical check [measured]: direct summation over the Farey graph gives $T_N(1) = 4.946$\(N\=\100),\4.986\(400),\4.997\(1600),\4.998\(3200), converging to $2\zeta(2)^2/\zeta(4) = 5$ exactly as the identity predicts $(2\cdot(\pi^2/6)^2/(\pi^4/90) = 5)$; at $\sigma = 0.75$ the computed values approach $2\zeta(1.5)^2/\zeta(3) \approx 11.35$. The check is confirmation, not proof; the identity is proved above.

3.3 What a global trace can preserve

Definition 3.3 (admissible global-trace identity class). Fix $\sigma>1/2$ and set $B:=4\zeta(2\sigma)$. An admissible identity consists of a fixed function $g:[0,B]\to\mathbb{R}$, independent of N, and fixed scalars $a\neq 0$, b, and c, such that

$$D(N)=a\ \mathrm{tr}\ g(L_N(\sigma))+b\ m_N+c$$

for all sufficiently large N. Allowing arbitrary N-dependent offsets would make the question vacuous.

Proposition 3.3.1 (uniform Farey counting and sublinear discrepancy) [proved]. Let `A_N(x):=#{p/q∈F_N:p/q≤x}`. Uniformly for `x∈[0,1]`,

$$A_N(x) = xm_N + O(N \log N). \tag{1}$$

Consequently,

$$\sup_{x\in[0,1]} \left|\frac{A_N(x)}{m_N} - x\right| = O\left(\frac{\log N}{N}\right). \tag{2}$$

and

$$D(N) = O(N \log N) = o(m_N). \tag{3}$$

Proof. For fixed q, Möbius inversion gives

$$\#\{1 \le p \le xq : \gcd(p, q) = 1\} = \sum_{d|q} \mu(d) \left\lfloor \frac{xq}{d} \right\rfloor = x\varphi(q) + O(\tau(q)). \tag{4}$$

uniformly in `x`. Summing over `q≤N`, and treating the two endpoints as an `O(1)` adjustment, yields

$$\begin{aligned} A_N(x) &= x\left(1 + \sum_{q\le N} \varphi(q)\right) + O\left(\sum_{q\le N} \tau(q)\right) \\ &= xm_N + O(N \log N). \end{aligned} \tag{5}$$

using the standard estimates `Σ_{q≤N}ϕ(q)=3N^2/π^2+O(N log N)` and `Σ_{q≤N}τ(q)=O(N log N)`. Since `m_N` is of order `N^2`, (2) follows. If `ρ_1<...<ρ_{m_N}` enumerates `F_N`, then `A_N(ρ_j)/m_N=j/m_N`; hence `|ρ_j-j/m_N|≤sup_x|A_N(x)/m_N-x|`. Summing over `j` proves (3). ■

Definition 3.3.2 (optional growth hypotheses, not findings). The following symbols are used only to state conditional corollaries:

- `H2`: `D(N)` is unbounded.
- `H3`: for every `ε>0`, `D(N)=Ω(N^{1/2-ε})` along a subsequence.

`H3` implies `H2`. Neither hypothesis is asserted as a theorem, measurement, or literature finding in this paper. The earlier labels `G2` and `G3` are preserved in the revision genealogy, but removing their fact-like presentation is part of this correction.

Theorem 3.4 (global-trace rigidity and conditional exclusions) [proved]. Fix `σ>1/2`, `B=4ζ(2σ)`, and `T(σ)=2ζ(2σ)^2/ζ(4σ)`.

(i) If g is Lipschitz on `[0,B]` with constant `K`, then

$$\operatorname{tr} g(L_N) = g(0)m_N + E_N, \qquad |E_N| \le KT(\sigma). \tag{6}$$

Therefore any admissible identity has either an extensive branch `D(N)=Θ(m_N)` or a cancelled branch `D(N)=O(1)`. Proposition 3.3.1 unconditionally excludes the extensive branch. Under the additional hypothesis `H2`, the cancelled branch is also excluded, so no admissible Lipschitz identity exists.

(ii) If |g(λ)-g(0)|$\le$K λ^α for $\alpha\in$(3/4,1), then

$$|E_N| \le C(K,\alpha,\sigma)m_N^{1-\alpha} = O\Big(N^{2(1-\alpha)}\Big). \tag{7}$$

Proposition 3.3.1 again excludes the extensive branch. Under H3, the cancelled branch is excluded after choosing 0<ε<1/2-2(1-α).

(iii) If g is merely continuous, then

$$\frac{\operatorname{tr} g(L_N)}{m_N} \longrightarrow g(0). \tag{8}$$

Thus Proposition 3.3.1 forces cancellation of the extensive coefficient, but without a quantitative modulus at zero the centered trace remains an open route.

Proof. For (i), positivity and Lemma 3.2.2 give

$$|E_N| \le K \sum_i \lambda_i(L_N) = KT_N(\sigma) \le KT(\sigma). \tag{9}$$

Substitution into Definition 3.3 gives the two branches, and the stated conclusions follow from Proposition 3.3.1 and, only for the second branch, the optional hypothesis H2.

For (ii), Corollary 3.2.3 gives N_N(t):=#{i:λ_i$\ge$t}$\le$min(m_N,T(σ)/t). The layer-cake formula yields

$$\sum_i \lambda_i^\alpha \le m_N t_*^\alpha + \frac{\alpha}{1-\alpha} T(\sigma) t_*^{\alpha-1}. \tag{10}$$

For all sufficiently large N, choosing t_*=T(σ)/m_N$\le$B proves (7); the finitely many smaller N are absorbed into the constant C. Under H3, its subsequential lower bound strictly dominates N^{2(1-α)} for the stated choice of ε.

For (iii), split the spectrum at t>0. Eigenvalues below t contribute at most m_N ω_g(t) and those above t number at most T(σ)/t. Divide by m_N, let N→∞, then let t→0. ∎

Remark 3.5 (scope). The unconditional theorem determines the asymptotic shape of fixed global traces. The full Lipschitz and Hölder no-go statements are conditional exactly where H2 or H3 is invoked. The result does not cover N-dependent test functions, shell or off-diagonal traces, rough continuous tests in the cancelled branch, exotic unit-weight domains, or distinguished-vector pairings.

Remark 3.6 (revision genealogy). Theorem 3.1 descends from the archived Ford-family packet, while Lemma 3.2.2 corrects the false trace-growth claims in drafts v1-v4. Version 2 adds the self-contained proof of sublinear Farey discrepancy and replaces the old fact-like G2-G3 presentation with optional H2-H3 hypotheses. No mathematical implication is lost; the evidentiary status is made exact.

4. Disposition of former measured exclusions

The v6 shell-diagonal exponent, single-mode Gauss-Kuzmin correlation, and continued-fraction-depth comparisons are not retained as results in this revision. The governed public package did not include the promised reruns, uncertainty estimates, or stable projector-based treatment of the nearly degenerate bottom eigenspace. Those observations remain private historical route material only. They prove no exclusion, support no theorem in §§2-3, and leave shell-diagonal, off-diagonal, higher-moment, and localization routes open.

A future measured companion would need pinned scripts and dependencies, machine-readable inputs and outputs, local exponent series, uncertainty estimates, and subspace-projector observables robust to eigenbasis choice. Until that package exists, this paper makes no numerical exclusion claim.

5. What survives

The elimination leaves two observable families, both developed in companion papers, and both evading the fence for the same structural reason: they are quadratic forms against *distinguished vectors*, not fixed global traces (so Theorem 3.4 does not see them). No retained shell-diagonal theorem or measurement excludes them.

(i) **Zero-mode pairings** (Papers 1 and 3) - the deficit $1 - \|L^{+}[U,L]\phi_0\|^2$ provably *equals* a Mertens-function pairing - an exact identity [proved, in the companion papers]; the RH-*equivalent* lower bound on it (the saturation law) remains open [conjectural]. No claim of RH is made.

(ii) **Cusp pairings** (Paper 7) - at the critical limit the rescaled family's spectral zeta is $\zeta(2w-1)/\zeta(2w)$ [proved, companion Paper 7, Theorem C-28; packets BC-016-fordfamily-zeta-norm-20260612, BC-018-critical-limit-totient-spectrum-20260612]. At the level proved this is the classical totient Dirichlet series $\Sigma\ \phi(q)\ q^{-2w} = \zeta(2w-1)/\zeta(2w)$ (Hardy & Wright), coinciding up to Γ-factors with the Eisenstein scattering ratio; the suggestion that the alignment of the existence boundary (Theorem 3.1), the rescaled limit, and this identity at $\mathrm{Re}\ s = 1/2$ anchors an interpolation program on $s \in (1/2,\ 1]$ is [conjectural]. We note that the trace identity $\mathrm{tr}\ L(\sigma) = 2\zeta(2\sigma)^2/\zeta(4\sigma)$ of Lemma 3.2.2 [proved] adds a second exact zeta-quotient evaluation attached to the same family.

Obstruction Observable	global trace tr g(L_N)	shell diagonal tr(M_q L^k)	zero-mode pairing $\langle\phi_0, U\phi_0\rangle$	cusp pairing
Thm 2.3 (unit weights)	blocked	blocked	blocked	blocked
Thm 3.1(b) ($\sigma \le 1/2$)	blocked	blocked	blocked	limit object only
Thm 3.4 ($\sigma > 1/2$)	extensive branch blocked unconditionally; cancelled branch blocked only under H2 or H3	not covered	**evades**	**evades**
Former §4 measurements	-	not retained; route open	untouched	untouched

The methodological moral: in this geometry, arithmetic lives in *distinguished vectors* (flat mode, cusp), not in global spectral averages.

6. Open problems

Q1 (the cancelled corner of Theorem 3.4). Fix $\sigma > 1/2$. Does there exist a continuous $g : [0, 4\zeta(2\sigma)] \to \mathbb{R}$, admitting no Hölder modulus of exponent $> 3/4$ at 0, and constants $a \neq 0$, c, with $a\,(\mathrm{tr}\ g(L\_N(\sigma)) - g(0)\ m\_N) + c = D(N)$ for all large N? Theorem 3.4(ii)-(iii) identifies this as the surviving unconditional configuration in the admissible class. A proof of nonexistence would close the class entirely; a construction would require the centered trace to reproduce the zero-driven oscillations of $D(N)$

from eigenvalue clustering at `0` alone, and the layer-cake bound shows it must exploit the regime `λ_i ∈ (0, T(σ)/m_N)` - eigenvalues of size `O(N^{-2})`.

Q2 (off-diagonal shell traces). Do the off-diagonal shell observables `tr(M_q L_N M_{q'} L_N)` (`q ≠ q'`), or diagonal traces of any fixed order, carry discrepancy information? The former finite probes are not retained in v7, so these route classes remain open. A governed measurement or theorem in either direction would move the fence.

Q3 (`N`-dependent observables). Definition 3.3 excludes `N`-dependent test functions `g_N` to avoid vacuity, but structured families (e.g. `g_N(λ) = g(m_N λ)` probing the `O(N^{-2})` spectral scale) are legitimate and unfenced. Formulate a non-vacuous admissible class of such families and prove a no-go or exhibit an identity. The trace identity of Lemma 3.2.2 suggests the natural normalization is `m_N λ`, since `Σ_i λ_i = T_N(σ) = O(1)` forces typical nonzero eigenvalues to scale like `1/m_N`.

Q4 (exotic domains at unit weights). Theorem 2.3 is a nonexistence result over the core `c_c(𝔉)`. Is there a dense subspace $\mathcal{D} \subset$ `ℓ²(𝔉)`, necessarily containing no finitely supported vector other than `0` (Remark 2.4), on which the unit-weight form is finite and closable? We expect not [conjectural], but the question is open.

Q5 (sharpness of the conditional Hölder threshold). Assuming `H3`, is the exponent `3/4` in Theorem 3.4(ii) sharp? The bound `|E_N| ≤ C m_N^{1-α}` is tight for the layer-cake argument given only the constraints `Σ λ_i ≤ T(σ)` and `λ_i ≤ B`; sharpness for the actual operators `L_N(σ)` would follow from a matching lower-bound construction using the true eigenvalue distribution near `0`, which is not currently known [conjectural].

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Archived proof-packet filenames remain in the private revision genealogy. They are not public theorem citations and are not premises required to verify the self-contained proofs in this manuscript.